Harbourline Capital · From document chase to investment decision

An AI-assisted, governed deal room on the Databricks Data Intelligence Platform

Private-markets investment · Deal origination, due diligence and post-investment monitoring

Harbourline Capital is a fictional Hong Kong private-markets investor; all data in this build is synthetic.

The outcome in one slide

	Today	With the AI deal room
Stage-gate cycle time (Stage 1 → 2)	~10 business days	**~4 days (-60 %)**
Analyst time on document triage	~15 min / document	**~3 min / document (-80 %)**
Cross-document figure checks	sampled, late (in IC pack prep)	**100 % of extracted figures, on arrival**
Quarterly portfolio pack	~3 weeks of consolidation	**~3 days, one certified number**
External-party document exchange	email attachments	**governed portal — the GP sees only its own uploads**

≈ 960 analyst hours a year back (≈ 0.5 FTE, ≈ US$ 90k) and investment-committee decisions about **6 days sooner per deal** — on one governed platform, no extra BI, workflow or AI tools.

Baseline and targets are planning assumptions for a fund house running ~40 active deals a year with ~60 diligence documents each (see "How we sized it").

Why this matters — two owners, one platform

Executive sponsor — COO / Head of Investment

- Faster, better-evidenced IC decisions: no stage advances until every required document **and** the named approval are in.
- Lower operational and conduct risk: one access model for people, dashboards, AI and outside parties; every action and AI call audited.
- Cost you can see: one platform bill instead of a document tool + BI tool + AI tool + workflow tool.

Domain owner — Deal-team lead / Investment operations

- The AI files, labels and reads every document, with a confidence score — the team reviews exceptions, not everything.
- Live completion %, gaps and risk alerts per deal room; no spreadsheet tracker.
- Self-service answers in English or Chinese (Genie) and AI-built report widgets on certified metrics — no ticket to the data team.

The problem today

1. **Documents arrive from many channels** — internal teams, the GP, email — and are classified and checked by hand.
2. **Completeness and approvals live in spreadsheets and inboxes**; nobody can see which deal is blocked and why.
3. **Figures disagree across documents** (e.g. FY revenue 5.4 vs 6.1 USD m) and are caught late — in IC pack preparation.
4. **KPIs are rebuilt for every report**; the dashboard, the board pack and ad-hoc answers do not always match.
5. **AI is blocked by governance**: who may see which portfolio, and what may an AI model read?

How we sized it (assumptions, to be validated in the pilot)

Driver	Assumption	Annual effect
Deal volume	40 active deals / year × 60 diligence documents	2,400 documents
Document triage	15 → 3 min / document (AI classifies + extracts; humans review < 0.8 confidence)	**≈ 480 h saved**
Quarterly monitoring pack	120 → 0 h of consolidation per quarter (certified metric views feed dashboard, Genie, reports)	**≈ 480 h saved**
Stage-gate wait	10 → 4 business days (live gaps, routed approvals, gate due-date alerts)	**≈ 6 days faster per deal**
Loaded analyst cost	US$ 95 / hour	**≈ US$ 90k / year**
Platform run cost (pilot scope)	serverless, pay per use; one shared SQL warehouse, scale-to-zero Lakebase	measured live in system tables

The solution — one integrated data journey

Stage	What runs	What the business gets
1 · Lakeflow — ingest	Auto Loader picks up every deal-room PDF from the landing zone (and optionally SharePoint); a daily Lakeflow job pulls public market factors (FRED + central-bank series)	Every document and market signal arrives without manual uploads
2 · Unity Catalog — govern	One role-scope table + one row-filter function + a tag-based ABAC policy; classification tags; lineage and audit	Define access once — the same rule protects the app, dashboards, Genie, AI and the GP portal
3 · Lakebase — operational serving	Postgres for the app's live state (gates, approvals, threads, proposals) + a synced table serving market factors from Delta	Millisecond reads and transactional approvals inside the app
4 · GenAI — make it intelligent	`ai_parse_document → ai_classify → ai_extract` with confidence; cross-document mismatch check; `ai_query` market-risk note; Widget Builder agent on an open-weight model	Documents filed, labelled and read automatically; risks surfaced with reasons
5 · Genie Agent — ask in natural language	Genie on certified metric views + a business glossary (EN / 繁 / 简)	"How did gross MOIC change this quarter?" — answered from the same definition as the dashboard
6 · Databricks App — surface it	The Harbourline Deal Room app: my deals, deal-room gate, AI file drawer, market board, Stage 3 dashboard, widget preview → approval, GP portal	One place for the deal team, the Head of Investment and the outside GP

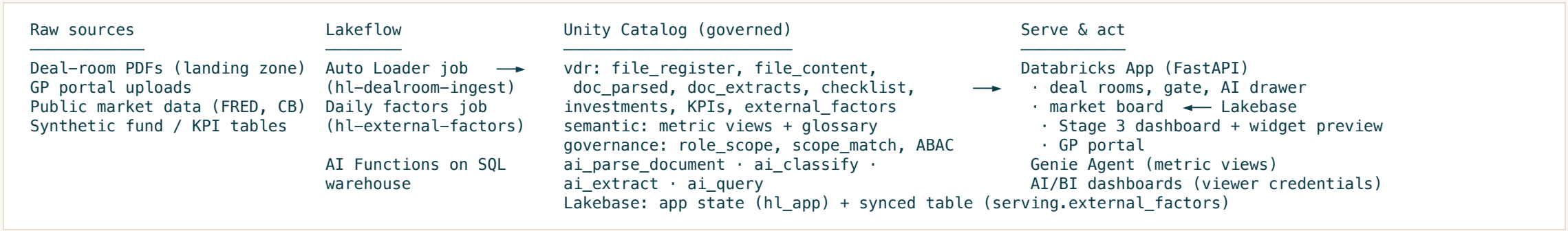
A day in the deal room (the demo storyline)

1. **Deal-team lead** opens *My deals → Solace Grid Energy*: 6 of 7 Stage 1 items done (85.7 %), locked on the Head of Investment approval.
2. Opens a file: the AI says *IC overview, 0.85* and *Confidential, 0.85* — auto-filed; a 0.78 score went to review.
3. **Head of Investment** sees **one more deal** than the Fund I owner (7 vs 6), approves, advances to Stage 2. The diligence pack lands; Auto Loader + AI file it; a **figure mismatch** alert appears.
4. The **market board** (served from Lakebase) and an **AI risk note** flag what moved for this deal.
5. On the Stage 3 dashboard and in **Genie**, the Fund I owner sees **5 companies**, the Head of Investment **6** — same dashboard, one policy. Genie answers *gross MOIC 1.36 → 1.39* with the certified measure.
6. The Fund I owner asks the **Widget Builder** for a KPI tile → live preview → the Head of Investment approves → it is published to the official dashboard.
7. The **external GP** logs in: sees **0** Harbourline documents, uploads its pack, sees only its own files.

Trust by design

- **Define access once, apply everywhere** — row filters on the portfolio tag decide what the app, AI/BI dashboard, Genie and the agent see.
  Evidence: the same query returns 6 companies for the Fund I owner, 7 for the Head of Investment, and 0 Harbourline files for the GP.
- **AI is advisory** — confidence scores on every classification and extraction; under 0.80 a person decides; nothing is approved by AI.
- **One audit trail** — every approval (person + role) in Lakebase; every query in system tables; lineage from file to dashboard tile.
- **Open and portable** — Delta tables, open-weight model, standard Postgres; no proprietary lock-in.

Architecture



Rollout plan

Weeks	Scope	Success criteria
1–2	Connect the real deal-room library and two live deals; map the 4-level classification	≥ 90 % of documents auto-filed at ≥ 0.80 confidence
3–4	Stage gates + approvals for one fund; Genie for the deal team	Stage 1 → 2 in ≤ 5 days on pilot deals; zero manual tracker
5–6	Quarterly monitoring pack from certified metrics; GP portal for one GP	Pack in ≤ 3 days; dashboard = Genie = report to the cent

Next steps

1. Agree the KPI baseline with the deal team (cycle time, triage minutes, pack effort) — two weeks of measurement.
2. Nominate the executive sponsor and a deal-team lead as domain owner.
3. Start the 6-week pilot on two live deals.